

3.4 Wireless Connectivity Options

Mobile phones are wireless gadgets, and the whole point is lost if you need to connect a cable to it for synchronising or transferring data! However, many mobile handsets boast of features that let you do this and more without the need to connect any cables. The following are the wireless connectivity options provided by mobile handsets today.

Infra-Red (IR)

The IrDA Standard, specified by the InfraRed Data Association (IrDA, formed in 1993), is intended to facilitate the point-to-point or point-to-multipoint communication between electronic devices (e.g. computers, mobile phones, peripherals) using directed infrared communications links through free space. Data transfer through infrared requires line-of-sight. Usually, each of the communicating devices has a transceiver (which is a combination of a receiver and a transmitter on one device). The data to be communicated is output from one device and converted to an analogue signal (infrared) and the other device receives it and converts it back to digital pulses.

Infrared communications are useful for indoor use. IR does not penetrate walls, and so does not interfere with devices in adjoining rooms.

The original IrDA standard, called IrDA 1.0, allowed for the transfer of data at up to 115.2 kbps at a range of up to 1 metre. In 1996, an extension to this standard called IrDA 1.1 was adopted, which allowed for data transfers up to 35 times faster than the original specification. This extension calls for data rates up to 4 Mbps, but retains compatibility with the original (115.2kbps) specification. Since it is relatively inexpensive to implement, most notebooks and handhelds as well as a large number of mobile phones are equipped with infrared ports for wireless data transfer.